

Machine Learning Project

Italian Rap Songs Dataset

Introduction to Machine Learning
March 2026 – May 2026

1 Introduction

The objective of this project is to apply the **Knowledge Discovery in Databases (KDD)** process and **Machine Learning techniques** to a real-world dataset of Italian rap songs.

2 Dataset Description

The dataset is composed of two CSV files.

2.1 File 1 – all_tracks.csv

Contains song-level information. Each row corresponds to one song.

Main groups of variables include:

Identifiers

- id – Unique identifier of the track
- id_artist – Artist identifier
- id_album – Album identifier

Song metadata

- title – Song title
- full_title – Full title
- album_name – Album name
- year, month, day – Release date
- duration_ms – Duration in milliseconds
- explicit – Explicit content flag

Lyrics information

- lyrics – Full lyrics
- language – Language of lyrics
- n_tokens – Number of words
- n_sentences – Number of sentences

- tokens_per_sent – Average words per sentence
- char_per_tok – Average characters per word
- lexical_density – Vocabulary richness

Swear features

- swear_IT – Italian swear words count
- swear_EN – English swear words count

Audio features

- bpm – Tempo
- centroid – Spectral centroid
- rolloff – Spectral rolloff
- flux – Spectral flux
- rms – Signal energy
- zcr – Zero crossing rate
- flatness – Spectral flatness
- spectral_complexity – Number of spectral peaks
- pitch – Fundamental frequency
- loudness – Perceived volume

Popularity indicators

- popularity – Popularity score
- stats_pageviews – Pageviews
- modified_popularity – Adjusted popularity indicator

2.2 File 2 – artists.csv

Contains artist-level information. Each row corresponds to one artist.

Variables include:

- id_author – Artist identifier
- name – Artist name
- gender – Gender
- birth_date – Date of birth
- birth_place – Birth location
- nationality – Nationality
- description – Biography

- active_start – Career start
- active_end – Career end
- province – Province
- region – Region
- country – Country
- latitude – Latitude
- longitude – Longitude

The datasets can be merged using:

```
all_tracks.id_artist = artists.id_author
```

3 Task 1 – Data Understanding and Preparation

This task follows the KDD process. Students must explore, clean, and prepare the dataset.

- Load both datasets, merge them, report number of songs, number of artists, number of features, distinguish between song level and artist level variables. Explain the role of the various audio, textual and geographic variables. All engineered features must also be documented.
- Evaluate missing values, duplicated records, inconsistent values, incorrect dates, extreme values (OOD).
- Analyse the distribution of popularity, pageviews, BPM, duration, lexical density, swear counts, release year. Recommended visualizations include histograms, boxplots, and violin plots.
- Compute correlation matrices. identify redundant variables, discuss meaningful relationships.
- Improve the dataset via feature engineering. Define, describe and motivate 3 features that could be added to the dataset computing them from what is available. Example: Swear density = $\frac{\text{swear_IT} + \text{swear_EN}}{\text{n_tokens}}$

4 Task 2 – Pattern Mining for Feature Discovery

In this task, students must apply the Apriori algorithm to discover patterns that will guide the creation of new predictive features.

The goal of this task is not only to extract association rules, but to use them to design meaningful variables for the predictive models developed later.

- Discretize selected numerical variables into categorical bins. Some examples: BPM → Slow / Medium / Fast. Popularity → Low / Medium / High. Duration → Short / Medium / Long. Swear density → Low / High. remember to justify your discretization strategy.
- Each song must be represented as a transaction composed of categorical items. For example: High BPM, Explicit, High Popularity. Encode the data as appropriate.

- Use Apriori to mine the data. Select minimum support and confidence, generate the frequent itemsets and extract association rules.
- Evaluate the rules in terms of support, confidence and lifts. Discuss which rules are interesting and meaningful.
- Transform selected rules into new binary features. For example, if a rule $\text{High BPM} \wedge \text{Explicit} \rightarrow \text{High Popularity}$ is discovered, a corresponding feature can be defined as $\text{is_fast_explicit} = \begin{cases} 1 & \text{if High BPM and Explicit} \\ 0 & \text{otherwise} \end{cases}$. Do this for at least 3 rules.

5 Task 3 – Unsupervised Learning

In this task, students must combine the engineered features and the rule-based features to cluster the data.

- Select a subset of relevant features for clustering. Consider audio features, textual features, engineered features, and rule-based binary features. Justify your selection.
- Preprocess the data appropriately. Apply normalization or standardization where needed, and handle categorical variables if present.
- Apply at least two clustering algorithms (e.g., K-Means, Hierarchical Clustering, DB-SCAN). Clearly describe the chosen methods and their parameters.
- Determine the optimal number of clusters when required (e.g., using the Elbow method, Silhouette score, or other criteria).
- Evaluate the quality of the clusters using internal validation metrics such as Silhouette score, Davies-Bouldin index, or similar measures.
- Interpret the clusters by analyzing their characteristics. Describe the profiles of each cluster in terms of audio, lyrics, and popularity features.
- Visualize the clustering results using dimensionality reduction techniques (e.g., PCA, t-SNE) and provide meaningful plots.

6 Task 4 – Feature Selection and Supervised Learning

In this task, students must combine the engineered features and the rule-based features to build predictive models.

- Define a good target by grouping macro-regions of Italy: North, Center, Campania, Lazio, Sardegna, South. Map the data accordingly and analyze target distribution.
- Analyze the usefulness of the available features. Use correlation analysis, variance filtering etc.
- Implement different classification models: decision trees, random forest, support vector machines. Remember to split the data in training and test as appropriate.
- Fine tune the models appropriately, trying different setups of features, for example analyzing importance and removing less important feature in subsequent trials
- Report accuracy, precision, recall, f1-score, confusion matrix.

7 Final Deliverables

Each team must submit:

1. Pdf Written Report
2. Source Code

With an email with the EXACT subject: [Group_id] IntroToMl First Project 2026 to both the following addresses: roberto.pellungrini@sns.it, fosca.giannotti@sns.it